

Derivation of State-Space Matrix for MPC Control

The following derivation follows the sequence and notation of the handwritten derivation. The factor of 2 is retained consistently in the front and rear lateral tire forces, and the required signs in the lateral-velocity and yaw-rate equations are corrected.

1 Vehicle Kinematic Equations

The vehicle states are the global longitudinal position X , lateral position Y , yaw angle ψ , longitudinal velocity v_x , lateral velocity v_y , and yaw rate r .

The global longitudinal-position equation is

$$\dot{X} = v_x \cos \psi - v_y \sin \psi. \quad (1)$$

The global lateral-position equation is

$$\dot{Y} = v_x \sin \psi + v_y \cos \psi. \quad (2)$$

The yaw-rate relation is

$$\dot{\psi} = r. \quad (3)$$

The longitudinal vehicle dynamics used in the handwritten model are

$$\dot{v}_x = a_{cmd} - \mu_{rr}g \cos \theta - g \sin \theta + v_y r. \quad (4)$$

Here, $\mu_{rr}g \cos \theta$ represents the rolling/frictional resistance, $g \sin \theta$ represents the road-grade contribution, and $v_y r$ is the body-frame coupling caused by yaw motion.

2 Front and Rear Normal Loads

The front normal force is

$$F_{zf} = \frac{mg \cos \theta a_2 - mh (\dot{v}_x + g \sin \theta)}{a_1 + a_2}. \quad (5)$$

Similarly, the rear normal force is

$$F_{zr} = \frac{mg \cos \theta a_1 + mh (\dot{v}_x + g \sin \theta)}{a_1 + a_2}. \quad (6)$$

For the nominal condition,

$$F_{zf,\text{nom}} = \frac{mga_2}{a_1 + a_2}, \quad (7)$$

and

$$F_{zr,\text{nom}} = \frac{mga_1}{a_1 + a_2}. \quad (8)$$

The front cornering stiffness is

$$C_{\alpha f} = C_{\alpha f,\text{nom}} \left(\frac{F_{zf}}{F_{zf,\text{nom}}} \right). \quad (9)$$

Similarly,

$$C_{\alpha r} = C_{\alpha r,\text{nom}} \left(\frac{F_{zr}}{F_{zr,\text{nom}}} \right). \quad (10)$$

The nominal values appearing in the handwritten derivation are

$$C_{\alpha f,\text{nom}} = 57,000 \text{ N/rad}, \quad (11)$$

and

$$C_{\alpha r,\text{nom}} = 52,000 \text{ N/rad}. \quad (12)$$

3 Front and Rear Slip Angles

The front slip angle is

$$\alpha_f = \delta - \frac{v_y + a_1 r}{v_x}. \quad (13)$$

The rear slip angle is

$$\alpha_r = -\frac{v_y - a_2 r}{v_x} = \frac{-v_y + a_2 r}{v_x}. \quad (14)$$

4 Lateral Tire Forces

Since the cornering stiffnesses are defined for individual tires, two tires contribute at each axle.

Therefore,

$$\boxed{F_{yf} = 2C_{\alpha f}\alpha_f}, \quad (15)$$

and

$$\boxed{F_{yr} = 2C_{\alpha r}\alpha_r}. \quad (16)$$

5 Newton–Euler Equations in the Vehicle-Fixed Frame

Applying Newton’s second law in the lateral direction,

$$\boxed{m(\dot{v}_y + v_x r) = F_{yf} + F_{yr}}. \quad (17)$$

Applying the yaw-moment equation about the center of gravity,

$$\boxed{I_z \dot{r} = a_1 F_{yf} - a_2 F_{yr}}. \quad (18)$$

5.1 Detailed Derivation of the Lateral-Velocity Equation

Starting from

$$m(\dot{v}_y + v_x r) = F_{yf} + F_{yr}, \quad (19)$$

substitute

$$F_{yf} = 2C_{\alpha f}\alpha_f, \quad F_{yr} = 2C_{\alpha r}\alpha_r. \quad (20)$$

Thus,

$$m(\dot{v}_y + v_x r) = 2C_{\alpha f}\alpha_f + 2C_{\alpha r}\alpha_r. \quad (21)$$

Expanding the left-hand side,

$$m\dot{v}_y + mv_x r = 2C_{\alpha f}\alpha_f + 2C_{\alpha r}\alpha_r. \quad (22)$$

Moving $mv_x r$ to the right-hand side,

$$m\dot{v}_y = 2C_{\alpha f}\alpha_f + 2C_{\alpha r}\alpha_r - mv_x r. \quad (23)$$

Dividing by m ,

$$\dot{v}_y = \frac{2C_{\alpha f}}{m}\alpha_f + \frac{2C_{\alpha r}}{m}\alpha_r - v_x r. \quad (24)$$

Now,

$$\alpha_f = \delta - \frac{v_y + a_1 r}{v_x}, \quad (25)$$

and

$$\alpha_r = \frac{-v_y + a_2 r}{v_x}. \quad (26)$$

Substituting these equations,

$$\begin{aligned} \dot{v}_y &= \frac{2C_{\alpha f}}{m} \left(\delta - \frac{v_y + a_1 r}{v_x} \right) \\ &\quad + \frac{2C_{\alpha r}}{m} \left(\frac{-v_y + a_2 r}{v_x} \right) - v_x r. \end{aligned} \quad (27)$$

Expanding the front term,

$$\begin{aligned} \frac{2C_{\alpha f}}{m} \left(\delta - \frac{v_y + a_1 r}{v_x} \right) &= \frac{2C_{\alpha f}}{m} \delta - \frac{2C_{\alpha f}}{mv_x} v_y \\ &\quad - \frac{2a_1 C_{\alpha f}}{mv_x} r. \end{aligned} \quad (28)$$

Expanding the rear term,

$$\frac{2C_{\alpha r}}{m} \left(\frac{-v_y + a_2 r}{v_x} \right) = -\frac{2C_{\alpha r}}{mv_x} v_y + \frac{2a_2 C_{\alpha r}}{mv_x} r. \quad (29)$$

Therefore,

$$\begin{aligned} \dot{v}_y &= \frac{2C_{\alpha f}}{m} \delta - \frac{2C_{\alpha f}}{mv_x} v_y - \frac{2a_1 C_{\alpha f}}{mv_x} r \\ &\quad - \frac{2C_{\alpha r}}{mv_x} v_y + \frac{2a_2 C_{\alpha r}}{mv_x} r - v_x r. \end{aligned} \quad (30)$$

Collecting the v_y terms,

$$\begin{aligned} & -\frac{2C_{\alpha f}}{mv_x} v_y - \frac{2C_{\alpha r}}{mv_x} v_y \\ &= -v_y \left(\frac{2C_{\alpha f} + 2C_{\alpha r}}{mv_x} \right). \end{aligned} \quad (31)$$

Collecting the r terms,

$$\begin{aligned} & -\frac{2a_1C_{\alpha f}}{mv_x}r + \frac{2a_2C_{\alpha r}}{mv_x}r - v_x r \\ & = r \left(\frac{-2a_1C_{\alpha f} + 2a_2C_{\alpha r}}{mv_x} - v_x \right). \end{aligned} \quad (32)$$

Therefore,

$$\dot{v}_y = -v_y \left(\frac{2C_{\alpha f} + 2C_{\alpha r}}{mv_x} \right) + r \left(\frac{-2a_1C_{\alpha f} + 2a_2C_{\alpha r}}{mv_x} - v_x \right) + \frac{2C_{\alpha f}}{m}\delta. \quad (33)$$

The front-axle yaw-rate contribution has a negative sign because the front slip-angle equation contains

$$-\frac{a_1 r}{v_x}. \quad (34)$$

5.2 Detailed Derivation of the Yaw-Rate Equation

Starting from

$$I_z \dot{r} = a_1 F_{yf} - a_2 F_{yr}, \quad (35)$$

substitute

$$F_{yf} = 2C_{\alpha f}\alpha_f, \quad (36)$$

and

$$F_{yr} = 2C_{\alpha r}\alpha_r. \quad (37)$$

Hence,

$$I_z \dot{r} = a_1 (2C_{\alpha f}\alpha_f) - a_2 (2C_{\alpha r}\alpha_r). \quad (38)$$

Therefore,

$$I_z \dot{r} = 2a_1 C_{\alpha f} \alpha_f - 2a_2 C_{\alpha r} \alpha_r. \quad (39)$$

Substituting the front and rear slip angles,

$$\begin{aligned} I_z \dot{r} &= 2a_1 C_{\alpha f} \left(\delta - \frac{v_y + a_1 r}{v_x} \right) \\ &\quad - 2a_2 C_{\alpha r} \left(\frac{-v_y + a_2 r}{v_x} \right). \end{aligned} \quad (40)$$

Expanding the front contribution,

$$2a_1C_{\alpha f} \left(\delta - \frac{v_y + a_1 r}{v_x} \right) = 2a_1C_{\alpha f}\delta - \frac{2a_1C_{\alpha f}}{v_x}v_y - \frac{2a_1^2C_{\alpha f}}{v_x}r. \quad (41)$$

Expanding the rear contribution,

$$-2a_2C_{\alpha r} \left(\frac{-v_y + a_2 r}{v_x} \right) = +\frac{2a_2C_{\alpha r}}{v_x}v_y - \frac{2a_2^2C_{\alpha r}}{v_x}r. \quad (42)$$

Hence,

$$\begin{aligned} I_z \dot{r} &= 2a_1C_{\alpha f}\delta - \frac{2a_1C_{\alpha f}}{v_x}v_y - \frac{2a_1^2C_{\alpha f}}{v_x}r \\ &\quad + \frac{2a_2C_{\alpha r}}{v_x}v_y - \frac{2a_2^2C_{\alpha r}}{v_x}r. \end{aligned} \quad (43)$$

Collecting the v_y terms,

$$-\frac{2a_1C_{\alpha f}}{v_x}v_y + \frac{2a_2C_{\alpha r}}{v_x}v_y = \frac{-2a_1C_{\alpha f} + 2a_2C_{\alpha r}}{v_x}v_y. \quad (44)$$

Collecting the r terms,

$$\begin{aligned} &-\frac{2a_1^2C_{\alpha f}}{v_x}r - \frac{2a_2^2C_{\alpha r}}{v_x}r \\ &= -\frac{2a_1^2C_{\alpha f} + 2a_2^2C_{\alpha r}}{v_x}r. \end{aligned} \quad (45)$$

Therefore,

$$\begin{aligned} I_z \dot{r} &= 2a_1C_{\alpha f}\delta + \frac{-2a_1C_{\alpha f} + 2a_2C_{\alpha r}}{v_x}v_y \\ &\quad - \frac{2a_1^2C_{\alpha f} + 2a_2^2C_{\alpha r}}{v_x}r. \end{aligned} \quad (46)$$

Dividing throughout by I_z ,

$$\boxed{\dot{r} = \frac{2a_1C_{\alpha f}}{I_z}\delta + \frac{-2a_1C_{\alpha f} + 2a_2C_{\alpha r}}{I_z v_x}v_y - \frac{2a_1^2C_{\alpha f} + 2a_2^2C_{\alpha r}}{I_z v_x}r.} \quad (47)$$

6 Continuous State-Space Representation

Using the structure shown in the handwritten derivation,

$$\dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u} + B_{ud}\mathbf{d}_{um} + B_{md}\mathbf{d}_m. \quad (48)$$

The output equation is

$$\mathbf{y} = C\mathbf{x} + D\mathbf{u}. \quad (49)$$

The state vector is

$$\mathbf{x} = \begin{bmatrix} X \\ Y \\ \psi \\ v_x \\ v_y \\ r \end{bmatrix}. \quad (50)$$

The control input is

$$\mathbf{u} = \begin{bmatrix} \delta \\ a_{cmd} \end{bmatrix}. \quad (51)$$

For linearization around an operating point $(\mathbf{x}_0, \mathbf{u}_0)$,

$$A = \left. \frac{\partial f}{\partial \mathbf{x}} \right|_{\mathbf{x}_0, \mathbf{u}_0}, \quad (52)$$

and

$$B = \left. \frac{\partial f}{\partial \mathbf{u}} \right|_{\mathbf{x}_0, \mathbf{u}_0}. \quad (53)$$

7 Derivation of the A Matrix

From

$$\dot{X} = v_x \cos \psi - v_y \sin \psi, \quad (54)$$

the nonzero partial derivatives are

$$\frac{\partial \dot{X}}{\partial \psi} = -v_x \sin \psi - v_y \cos \psi, \quad (55)$$

$$\frac{\partial \dot{X}}{\partial v_x} = \cos \psi, \quad (56)$$

and

$$\frac{\partial \dot{X}}{\partial v_y} = -\sin \psi. \quad (57)$$

Hence,

$$A_{13} = -v_{x0} \sin \psi_0 - v_{y0} \cos \psi_0, \quad (58)$$

$$A_{14} = \cos \psi_0, \quad (59)$$

and

$$A_{15} = -\sin \psi_0. \quad (60)$$

From

$$\dot{Y} = v_x \sin \psi + v_y \cos \psi, \quad (61)$$

we obtain

$$\frac{\partial \dot{Y}}{\partial \psi} = v_x \cos \psi - v_y \sin \psi, \quad (62)$$

$$\frac{\partial \dot{Y}}{\partial v_x} = \sin \psi, \quad (63)$$

and

$$\frac{\partial \dot{Y}}{\partial v_y} = \cos \psi. \quad (64)$$

Hence,

$$A_{23} = v_{x0} \cos \psi_0 - v_{y0} \sin \psi_0, \quad (65)$$

$$A_{24} = \sin \psi_0, \quad (66)$$

and

$$A_{25} = \cos \psi_0. \quad (67)$$

Since

$$\dot{\psi} = r, \quad (68)$$

we obtain

$$\boxed{A_{36} = 1}. \quad (69)$$

For

$$\dot{v}_x = a_{cmd} - \mu_{rr}g \cos \theta - g \sin \theta + v_y r, \quad (70)$$

the relevant state derivatives are

$$\frac{\partial \dot{v}_x}{\partial v_y} = r, \quad (71)$$

and

$$\frac{\partial \dot{v}_x}{\partial r} = v_y. \quad (72)$$

Thus,

$$A_{45} = r_0, \quad (73)$$

and

$$A_{46} = v_{y0}. \quad (74)$$

For the lateral equation,

$$\begin{aligned} \dot{v}_y = & -\frac{2C_{\alpha f} + 2C_{\alpha r}}{mv_x}v_y \\ & + \left(\frac{-2a_1C_{\alpha f} + 2a_2C_{\alpha r}}{mv_x} - v_x \right) r + \frac{2C_{\alpha f}}{m}\delta. \end{aligned} \quad (75)$$

Differentiating with respect to v_x ,

$$\begin{aligned} \frac{\partial \dot{v}_y}{\partial v_x} = & \frac{(2C_{\alpha f} + 2C_{\alpha r})v_y}{mv_x^2} \\ & - \frac{(-2a_1C_{\alpha f} + 2a_2C_{\alpha r})r}{mv_x^2} - r. \end{aligned} \quad (76)$$

Therefore,

$$A_{54} = \frac{(2C_{\alpha f} + 2C_{\alpha r})v_{y0} - (-2a_1C_{\alpha f} + 2a_2C_{\alpha r})r_0}{mv_{x0}^2} - r_0. \quad (77)$$

Differentiating with respect to v_y ,

$$A_{55} = -\frac{2C_{\alpha f} + 2C_{\alpha r}}{mv_{x0}}. \quad (78)$$

Differentiating with respect to r ,

$$A_{56} = \frac{-2a_1C_{\alpha f} + 2a_2C_{\alpha r}}{mv_{x0}} - v_{x0}. \quad (79)$$

For the yaw-rate equation,

$$\begin{aligned} \dot{r} = & \frac{-2a_1C_{\alpha f} + 2a_2C_{\alpha r}}{I_z v_x} v_y \\ & - \frac{2a_1^2C_{\alpha f} + 2a_2^2C_{\alpha r}}{I_z v_x} r + \frac{2a_1C_{\alpha f}}{I_z} \delta. \end{aligned} \quad (80)$$

Differentiating with respect to v_x ,

$$\begin{aligned} \frac{\partial \dot{r}}{\partial v_x} = & -\frac{(-2a_1C_{\alpha f} + 2a_2C_{\alpha r})v_y}{I_z v_x^2} \\ & + \frac{(2a_1^2C_{\alpha f} + 2a_2^2C_{\alpha r})r}{I_z v_x^2}. \end{aligned} \quad (81)$$

Thus,

$$A_{64} = \frac{-(-2a_1C_{\alpha f} + 2a_2C_{\alpha r})v_{y0} + (2a_1^2C_{\alpha f} + 2a_2^2C_{\alpha r})r_0}{I_z v_{x0}^2}. \quad (82)$$

Differentiating with respect to v_y ,

$$A_{65} = \frac{-2a_1C_{\alpha f} + 2a_2C_{\alpha r}}{I_z v_{x0}}. \quad (83)$$

Differentiating with respect to r ,

$$A_{66} = -\frac{2a_1^2C_{\alpha f} + 2a_2^2C_{\alpha r}}{I_z v_{x0}}. \quad (84)$$

Therefore,

$$A = \begin{bmatrix} 0 & 0 & -v_{x0} \sin \psi_0 - v_{y0} \cos \psi_0 & \cos \psi_0 & -\sin \psi_0 & 0 \\ 0 & 0 & v_{x0} \cos \psi_0 - v_{y0} \sin \psi_0 & \sin \psi_0 & \cos \psi_0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & r_0 & v_{y0} \\ 0 & 0 & 0 & A_{54} & -\frac{2C_{\alpha f} + 2C_{\alpha r}}{mv_{x0}} & \frac{-2a_1 C_{\alpha f} + 2a_2 C_{\alpha r}}{mv_{x0}} - v_{x0} \\ 0 & 0 & 0 & A_{64} & \frac{-2a_1 C_{\alpha f} + 2a_2 C_{\alpha r}}{I_z v_{x0}} & -\frac{2a_1^2 C_{\alpha f} + 2a_2^2 C_{\alpha r}}{I_z v_{x0}} \end{bmatrix}. \quad (85)$$

8 Derivation of the B Matrix

The input vector is

$$\mathbf{u} = \begin{bmatrix} \delta \\ a_{cmd} \end{bmatrix}. \quad (86)$$

The steering angle δ enters the lateral-velocity equation through

$$\frac{2C_{\alpha f}}{m} \delta. \quad (87)$$

Therefore,

$$\frac{\partial \dot{v}_y}{\partial \delta} = \frac{2C_{\alpha f}}{m}. \quad (88)$$

Similarly, the steering angle enters the yaw-rate equation through

$$\frac{2a_1 C_{\alpha f}}{I_z} \delta. \quad (89)$$

Therefore,

$$\frac{\partial \dot{r}}{\partial \delta} = \frac{2a_1 C_{\alpha f}}{I_z}. \quad (90)$$

The longitudinal acceleration command appears in

$$\dot{v}_x = a_{cmd} - \mu_{rr} g \cos \theta - g \sin \theta + v_y r. \quad (91)$$

Therefore,

$$\frac{\partial \dot{v}_x}{\partial a_{cmd}} = 1. \quad (92)$$

Hence,

$$B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 1 \\ \frac{2C_{\alpha f}}{m} & 0 \\ \frac{2a_1 C_{\alpha f}}{I_z} & 0 \end{bmatrix}. \quad (93)$$

9 Unmodelled-Force Disturbance Matrix

The unmodelled-force disturbance vector is

$$\mathbf{d}_{um} = \begin{bmatrix} \Delta F_{x,\text{frict}} \\ \Delta F_{yf} \\ \Delta F_{yr} \end{bmatrix}. \quad (94)$$

The longitudinal-force disturbance contributes

$$\Delta \dot{v}_x = \frac{\Delta F_{x,\text{frict}}}{m}. \quad (95)$$

The lateral-force disturbances contribute

$$\Delta \dot{v}_y = \frac{\Delta F_{yf} + \Delta F_{yr}}{m}. \quad (96)$$

The yaw-rate disturbance contribution is

$$\Delta \dot{r} = \frac{a_1 \Delta F_{yf} - a_2 \Delta F_{yr}}{I_z}. \quad (97)$$

Therefore,

$$B_{ud} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ \frac{1}{m} & 0 & 0 \\ 0 & \frac{1}{m} & \frac{1}{m} \\ 0 & \frac{a_1}{I_z} & -\frac{a_2}{I_z} \end{bmatrix}. \quad (98)$$

The modeled disturbance vector in the handwritten notes is

$$\mathbf{d}_m = \begin{bmatrix} \delta_{\text{user}} \\ \theta_{\text{grade}} \end{bmatrix}. \quad (99)$$

Accordingly, B_{md} represents the corresponding modeled-disturbance input matrix.

10 Output Equation

The output matrix selects X , Y , and ψ :

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}. \quad (100)$$

For zero direct feedthrough,

$$D = 0. \quad (101)$$

Hence,

$$\mathbf{y} = C\mathbf{x} + D\mathbf{u} = C\mathbf{x}. \quad (102)$$

11 Final Corrected Continuous Model

The complete corrected continuous vehicle model is

$$\begin{aligned} \dot{X} &= v_x \cos \psi - v_y \sin \psi, \\ \dot{Y} &= v_x \sin \psi + v_y \cos \psi, \\ \dot{\psi} &= r, \\ \dot{v}_x &= a_{cmd} - \mu_{rr} g \cos \theta - g \sin \theta + v_y r, \\ \dot{v}_y &= -v_y \left(\frac{2C_{\alpha f} + 2C_{\alpha r}}{mv_x} \right) + r \left(\frac{-2a_1 C_{\alpha f} + 2a_2 C_{\alpha r}}{mv_x} - v_x \right) + \frac{2C_{\alpha f}}{m} \delta, \\ \dot{r} &= \frac{-2a_1 C_{\alpha f} + 2a_2 C_{\alpha r}}{I_z v_x} v_y - \frac{2a_1^2 C_{\alpha f} + 2a_2^2 C_{\alpha r}}{I_z v_x} r + \frac{2a_1 C_{\alpha f}}{I_z} \delta. \end{aligned} \quad (103)$$